

Truncated Divisor Bands and the Reciprocal-Emitter Boundary Operator

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Abstract

The preceding TPC-211 analysis gives an exact logarithmic Möbius derivative for a complete squarefree divisor packet of the literal product-coupled Euler profiles. The physical transition, however, uses the non-Boolean band $Y_0 < d \leq U$ and a divisor-dependent reciprocal emitter. This paper isolates that boundary interface. For any selected divisor family we prove that its common-endpoint leakage is the signed incidence vector of the selected Boolean faces, with coefficients in the independent basis $(\log p)_{p \in \mathcal{P}}$. The selected packet is exactly the complete packet minus the missing-subset boundary. We then define a finite reciprocal occupancy operator and prove its exact collision Gram identity. In the natural direct sum over divisor residue spaces the emitter Gram is block diagonal and full rank; normalized residuals attain coherent-to-diagonal ratios equal to the number of divisor blocks. Thus a divisor cut and reciprocal occupancy, without a theorem coupling the literal residual profiles across divisors, do not force a saving. The finite checks use exact rational arithmetic and a unit-weight reciprocal fixture ($\psi = 1$); they are structural certificates, not asymptotic prime-distribution evidence. The physical smooth emitter, prime shell, cross-divisor profile theorem, Gate B, and the twin-prime endpoint remain open.

Claim level. PROVED_STRUCTURAL_L1 / STOP_SCOPED

BOUNDARY_EMITTER. No arithmetic L^2 credit or fixed-atom credit is claimed.

1 Introduction

The working twin-prime route in this repository studies a fixed gap through a literal transition carrier. Its central difficulty is not a missing formal Fourier identity but the simultaneous preservation of the divisor signs, the physical zero axis, the prime shell, and the cross-divisor Gram structure. The local V46 compiler writes the transition residual in the form

$$\mathfrak{R}_x^{\text{AP}} = -H \sum_{Y_0 < d \leq U} \sum_{r \bmod d} A_d(r) \widehat{\mathcal{R}}_d(r), \quad (1)$$

where, schematically, as in the V46 transition artifact [3],

$$A_d(r) = \frac{\mu(d) \log d}{d} \sum_{q \in \mathcal{Q}} \sum_{0 < |m| \leq dq/H} \psi\left(\frac{Hm}{dq}\right) \mathbf{1}_{r \equiv m\bar{q} \pmod{d}}. \quad (2)$$

The residue vector $\mathcal{R}_d$ is the literal physical profile at divisor d .

TPC-209 showed that a whole-frame Poisson transform leaves divisor-dependent multiplicative permutations. TPC-210 showed that independent admissible Poisson profiles can realize a Möbius-aligned Gram obstruction [1]. TPC-211 then moved to the literal product-coupled Euler profiles. It proved a product cocycle, full divisor rank, and the exact complete-packet identity [2].

$$\sum_{\emptyset \neq S \subseteq \mathcal{P}} \mu(d_S) \log(d_S) \Delta_S = - \sum_{p \in \mathcal{P}} \log p \mathcal{D}_p, \quad (3)$$

where $d_S = \prod_{p \in S} p$ and Δ_S is the product-profile defect. The common endpoint cancels on the complete packet when $|\mathcal{P}| \geq 2$.

The identity in (3) does not apply directly to (1). The active divisor set is a band in the divisor lattice, not a complete packet, and A_d changes with d . The precise question of this paper is therefore:

Does the actual divisor cut together with the reciprocal map force a cross-divisor cancellation before one proves a new theorem for the literal physical profiles?

We answer the algebraic question negatively, but in a controlled way. The answer has two parts. First, the cut produces an exact endpoint leakage vector. Second, the reciprocal map has an exact collision Gram, but its natural direct-sum geometry has no cross-divisor entries. This does not disprove a future estimate on the coupled physical family. It identifies the missing input: a coupling theorem must relate $\mathcal{R}_d$ across divisors after the boundary and emitter are retained.

Our contributions are:

1. an exact signed-incidence formula for any divisor selector and an exact complete-minus-missing decomposition;
2. an exact reciprocal occupancy collision identity and a block-diagonal direct-sum Gram theorem;
3. a sharp finite alignment construction showing that emitter geometry alone cannot supply a universal cross-divisor saving;
4. exact finite certificates covering four cuts, 5,810 profile coordinates, and nine emitter divisor rows.

Scope firewall. The proof is finite and algebraic. We do not estimate the shifted-prime sequence, the hybrid profile, the smooth weight ψ , the prime shell, the four-packet reassembly, or any asymptotic twin-prime count. The unit-weight emitter used in the certificate is explicitly a modeling fixture.

2 The divisor band as a signed Boolean boundary

2.1 Notation

Let

$$\mathcal{P} = \{p_1, \dots, p_s\}, \quad p_i > 2,$$

be a finite set of distinct active primes. For a nonempty subset $S \subseteq \mathcal{P}$, put

$$d_S = \prod_{p \in S} p, \quad \varepsilon_S = (-1)^{|S|}.$$

The squarefree divisors of $t = \prod_{p \in \mathcal{P}} p$ are indexed by the nonempty faces of the Boolean cube. A divisor band is represented by a selector

$$\mathcal{A} \subseteq 2^{\mathcal{P}} \setminus \{\emptyset\}.$$

For the physical interval, the selector is

$$\mathcal{A}_{Y,U}(t) = \{S : Y_0 < d_S \leq U\}. \quad (4)$$

The complement within the nonempty Boolean lattice is denoted by

$$\mathcal{M} = (2^{\mathcal{P}} \setminus \{\emptyset\}) \setminus \mathcal{A}.$$

For every active prime define the signed incidence

$$\eta_p(\mathcal{A}) = \sum_{\substack{S \in \mathcal{A} \\ p \in S}} \varepsilon_S. \quad (5)$$

The corresponding logarithmic endpoint coefficient is

$$L(\mathcal{A}) = \sum_{S \in \mathcal{A}} \varepsilon_S \log d_S = \sum_{p \in \mathcal{P}} \eta_p(\mathcal{A}) \log p. \quad (6)$$

Lemma 2.1 (complete-packet incidence cancellation). *If $s \geq 2$, then*

$$\eta_p(2^{\mathcal{P}} \setminus \{\emptyset\}) = 0 \quad (p \in \mathcal{P}).$$

Consequently, the complete packet has $L = 0$.

Proof. Fix p . Pair each subset S containing p with the subset obtained by adding or removing one fixed prime $r \neq p$. The two signs are opposite. Equivalently,

$$\eta_p = (-1) \sum_{T \subseteq \mathcal{P} \setminus \{p\}} (-1)^{|T|} = -(1 - 1)^{s-1} = 0.$$

The logarithmic statement follows from (6). □

Theorem 2.2 (exact endpoint leakage). *For every selector $\mathcal{A}$, the common endpoint contribution of the weighted packet is exactly $L(\mathcal{A})$ times that endpoint. If $\mathcal{A}$ is not complete and $\eta_p(\mathcal{A}) \neq 0$ for some p , then the endpoint contribution cannot vanish uniformly over endpoint values. More precisely, for a scalar endpoint w ,*

$$\sum_{S \in \mathcal{A}} \varepsilon_S \log(d_S) w = L(\mathcal{A}) w.$$

Moreover, $L(\mathcal{A}) = 0$ if and only if $\eta_p(\mathcal{A}) = 0$ for every p .

Proof. The first identity is linearity and (6). Suppose $L(\mathcal{A}) = 0$. Then

$$\prod_{p \in \mathcal{P}} p^{\eta_p(\mathcal{A})} = 1$$

after exponentiating the logarithmic relation. The exponents are integers, so unique factorization gives $\eta_p(\mathcal{A}) = 0$ for all p . The converse is immediate from (6). If some incidence is nonzero then the logarithmic coefficient is nonzero, and choosing $w \neq 0$ gives nonzero leakage. □

Example 2.3 (the smallest physical band witness). Take $t = 5 \cdot 7 = 35$ and $Y_0 = 5$, $U = 35$. Then

$$\mathcal{A}_{5,35}(35) = \{\{7\}, \{5, 7\}\}.$$

The signs are $-1, +1$, so

$$\eta_5 = 1, \quad \eta_7 = 0, \quad L(\mathcal{A}) = -\log 7 + \log 35 = \log 5.$$

Thus the first proper divisor cut already restores a common endpoint term. This is an exact witness, not a numerical approximation to a logarithm.

2.2 Complete minus missing is an operator identity

Let Δ_S be any family of vectors in a common real or complex vector space and let w be a common endpoint vector. Write

$$R_S = w - \Delta_S, \quad \mathcal{T}_{\mathcal{A}} = \sum_{S \in \mathcal{A}} \varepsilon_S \log(d_S) R_S.$$

The next statement does not assume the product formula for Δ_S .

Proposition 2.4 (boundary decomposition). With $\mathcal{M}$ the missing subset family,

$$\mathcal{T}_{\mathcal{A}} = \mathcal{T}_{\text{full}} - \mathcal{T}_{\mathcal{M}}.$$

If the complete packet is a TPC-211 product packet with at least two active primes, then its endpoint-free part is the marked-prime derivative, while the missing packet contributes the exact boundary

$$\mathcal{T}_{\mathcal{M}} = L(\mathcal{M})w - \sum_{S \in \mathcal{M}} \varepsilon_S \log(d_S) \Delta_S.$$

The endpoint coefficient of the selected packet is $L(\mathcal{A}) = -L(\mathcal{M})$.

Proof. The first identity is the partition of the complete nonempty subset family into $\mathcal{A}$ and $\mathcal{M}$. Expanding $R_S = w - \Delta_S$ gives

$$\mathcal{T}_{\mathcal{M}} = L(\mathcal{M})w - \sum_{S \in \mathcal{M}} \varepsilon_S \log(d_S) \Delta_S.$$

Lemma 2.1 gives $L(\mathcal{A}) + L(\mathcal{M}) = 0$ when the full packet has at least two active primes. No estimate or limiting argument is used. $\square$

Remark 2.5. The boundary in (2.4) has two distinct pieces: endpoint leakage and missing profile mass. Treating the first complete-packet identity as if it bounded both pieces silently changes the divisor set. The physical emitter adds a third issue because its row also changes with d .

3 The reciprocal occupancy operator

3.1 Finite emitter model

Fix a positive integer H , a divisor $d \geq 2$, and a finite set $\mathcal{Q}$ of integers coprime to d . Define the finite index set

$$I_d = \left\{ (q, m) : q \in \mathcal{Q}, 0 < |m| \leq \left\lfloor \frac{dq}{H} \right\rfloor \right\}.$$

For unit weights, define the occupancy operator

$$E_d : \mathbb{C}^{I_d} \longrightarrow \mathbb{C}^{\mathbb{Z}/d\mathbb{Z}}, \quad (E_d a)(r) = \sum_{(q,m) \in I_d} a(q,m) \mathbf{1}_{r \equiv m\bar{q} \pmod{d}}.$$

The physical emitter is the same pushforward with

$$a(q,m) = \psi\left(\frac{Hm}{dq}\right),$$

followed by the nonzero scalar $\mu(d) \log(d)/d$. We first isolate the pushforward algebra.

Lemma 3.1 (reciprocal collision identity). *Let $a \equiv 1$ on I_d and write $N_d = E_d a$. Then*

$$\|N_d\|_2^2 = \sum_{(q_1, m_1), (q_2, m_2) \in I_d} \mathbf{1}_{d|m_1 q_2 - m_2 q_1}. \quad (7)$$

More generally, for arbitrary weights a ,

$$\|E_d a\|_2^2 = \sum_{I_d \times I_d} a(q_1, m_1) \overline{a(q_2, m_2)} \mathbf{1}_{d|m_1 q_2 - m_2 q_1}. \quad (8)$$

Proof. Expand the squared norm and use

$$m_1 \bar{q}_1 \equiv m_2 \bar{q}_2 \pmod{d}.$$

Multiplication by $q_1 q_2$, which is invertible modulo d , converts this condition exactly into $m_1 q_2 \equiv m_2 q_1 \pmod{d}$. The unit-weight case is the specialization $a = 1$. $\square$

Proposition 3.2 (direct-sum emitter Gram). *Let $\mathcal{B}$ be a finite divisor family and set*

$$\mathcal{H}_{\mathcal{B}} = \bigoplus_{d \in \mathcal{B}} \mathbb{C}^{\mathbb{Z}/d\mathbb{Z}}.$$

Let v_d be the unit-weight occupancy vector in the d -block, embedded in this direct sum. Then

$$\langle v_d, v_e \rangle = 0 \quad (d \neq e), \quad \langle v_d, v_d \rangle = \|v_d\|_2^2.$$

If every v_d is nonzero, the Gram matrix of the family has rank $|\mathcal{B}|$. For any signs $\sigma_d \in \{\pm 1\}$ the residuals

$$r_d = \sigma_d \frac{v_d}{\|v_d\|_2^2}$$

satisfy

$$\sigma_d \langle v_d, r_d \rangle = 1.$$

Consequently the coherent-to-diagonal ratio of these unit contributions is $|\mathcal{B}|$.

Proof. Different direct-sum blocks are orthogonal, proving (3.2). Nonzero diagonal entries give full rank. Equation (3.2) gives $\langle v_d, r_d \rangle = \sigma_d$, hence (3.2). The coherent energy is $|\mathcal{B}|^2$ and the diagonal energy is $|\mathcal{B}|$. $\square$

Remark 3.3. Proposition 3.2 is deliberately a negative interface result. It does not say that the literal residual family can choose the aligned residuals independently. It says that the reciprocal pushforward itself has not supplied the missing coupling. A saving theorem must use extra structure of $\mathcal{R}_d$, not only the congruence defining E_d .

Table 1: Boundary-band certificate cases. Incidence is ordered by the active prime list.

$\mathcal{P}$	Y_0	U	active divisors	$\eta(\mathcal{A})$	coordinates
(5, 7)	5	35	7, 35	(1, 0)	35
(5, 7, 11)	5	35	7, 35, 11	(1, 0, -1)	385
(5, 7, 11)	10	77	35, 11, 55, 77	(2, 2, 1)	385
(5, 7, 11, 13)	30	100	35, 55, 77, 65, 91	(3, 3, 2, 2)	5005

Table 2: Unit-weight reciprocal-emitter fixtures. The displayed Gram rank is the rank in the natural direct sum over the listed divisor blocks.

divisors	$\mathcal{Q}$	H	$\ v_d\ _2^2$	rank	ratio
(7, 35)	(2, 3, 13)	10	(86, 470)	2	2
(7, 11, 35, 55)	(2, 3, 13)	10	(86, 158, 470, 710)	4	4
(35, 55, 77)	(2, 3, 13)	10	(470, 710, 1014)	3	3

4 Exact finite certificates

4.1 Protocol

The certificate in the project directory is generated by exact rational code. The boundary test uses deterministic rational profiles on the common CRT coordinate set of size $\prod p$. This makes the equality in (2.4) a coefficientwise finite identity rather than a floating-point regression. The emitter test uses the finite unit-weight fixture $\psi = 1$. For each divisor the code records the occupancy vector, its squared norm, and the independent ordered collision count in (7).

The independent checker reimplements the Boolean and reciprocal arithmetic; it does not import the producer module. The optimized Python run is included as a second execution mode, not as an independent mathematical proof.

The first row is the exact witness in Example 2.3. The other rows show that a divisor band can leave several nonzero incidence coordinates; the effect is not peculiar to a two-prime packet. In all four rows the full incidence vector is zero and the selected plus missing incidence vectors add to it exactly.

The collision counts equal the squared norms entry by entry. The residuals in (3.2), with the literal Möbius signs of the displayed divisors, make all weighted contributions equal to one. This is a sharp finite alignment statement for the emitter interface.

4.2 Certificate status

The exact certificate covers four boundary cases, 5,810 profile coordinates, three emitter cases, and nine divisor rows. The producer, independent checker, and optimized independent checker all pass. The following status is part of the release contract:

```

TPC212_ROUTE_ADVANCE = YES
TPC212_STRUCTURAL_THRESHOLD_A = PASS
TPC212_CUT_ENDPOINT_LEAKAGE = PROVED_EXACT
TPC212_BOUNDARY_DECOMPOSITION = PROVED_EXACT
TPC212_RECIPROCAL_COLLISION = PROVED_EXACT_FINITE
TPC212_EMITTER_GRAM = PROVED_EXACT_BLOCK_DIAGONAL

```

```

TPC212_EMITTER_ONLY_UNIVERSAL_SAVING = REFUTED_SCOPED
TPC212_LITERAL_PHYSICAL_BOUNDARY_BOUND = OPEN
TPC212_PHYSICAL_CROSS_DIVISOR_GRAM_BOUND = OPEN
TPC212_ARITHMETIC_ADVANCE = NO
TPC212_FIXED_ATOM_CREDIT = 0
TPC212_L2 = NONE
TPC212_FULL_GATE_B_STRICT_1_OVER_400 = UNPAID

```

5 What the obstruction does and does not say

5.1 The missing physical coupling

The direct sum in Proposition 3.2 is the correct ambient space for an arbitrary family of residue vectors $(\mathcal{R}_d)_{d \in \mathcal{B}}$. It is also exactly the space in which a separate Cauchy estimate sees one emitter row per divisor. The proposition shows why that estimate cannot be upgraded by rhetoric about “the same reciprocal variable”: after the residue pushforward, different divisor blocks are still independent unless a new map identifies them.

For the literal TPC object, such an identifying map might come from the product-coupled Euler factors, the common endpoint, or the common underlying integer u . TPC-211 proved that product rank and a shared endpoint do not provide it automatically. TPC-212 adds that the cut and reciprocal occupancy do not provide it either. The remaining theorem must therefore be of the form

$$\left| \sum_{d \in \mathcal{A}_{Y,U}(t)} \frac{\mu(d) \log d}{d} \langle E_d \psi_d, \widehat{\mathcal{R}}_d \rangle \right| \leq \text{a saving bound for the coupled family}, \quad (9)$$

where the right side must use a relation between the different $\widehat{\mathcal{R}}_d$. Applying Cauchy in the direct sum, replacing $\widehat{\mathcal{R}}_d$ by arbitrary unit vectors, or replacing the smooth ψ_d by independent unit weights cannot prove such a relation.

5.2 Why the finite fixture is not an arithmetic result

The finite emitter certificate sets $\psi = 1$ on a finite set of (q, m) pairs. It is useful because the congruence collision is then an integer identity and because every nonzero block can be aligned explicitly. It is not the physical smooth emitter, does not include the prime shell, and does not control the relative phases of the literal residuals. Therefore the correct label is `NUMERICALLY_CERTIFIED` for the finite rows and `MODELING_CHOICE` for the unit-weight fixture, not an arithmetic theorem.

The endpoint incidence theorem is stronger in a different direction: it uses the exact logarithmic basis and unique factorization, so it is a general finite identity for every cut selector. Still, it only identifies the term that a future physical estimate must pay; it does not bound that term in the asymptotic transition range.

6 Conclusion and next route

TPC-212 converts the open phrase “truncated boundary plus reciprocal emitter” into two explicit operators. The divisor cut has a signed Boolean incidence vector, and the reciprocal map has a collision Gram. The complete packet annihilates the first vector, while a proper band generally does not. The natural emitter Gram is block diagonal and admits sharp finite alignment.

The strongest positive result is the exact boundary/operator decomposition. The strongest obstruction is the scoped failure of emitter-only universal saving. The next theorem is now sharply specified: construct and bound the coupling map that takes the literal V46 profile family into these divisor blocks, retaining the smooth weights and the prime shell before any outer absolute value. Until that theorem is proved, Gate B remains open and no twin-prime conclusion follows.

References

- [1] Liang Wang. Poisson profile realizability and the mobius alignment obstruction, 2026. TPC-210 project release in the prime dynamics theory repository.
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